MoBE-BCN: Mixture of Bilinear Experts for Sample-Efficient Algebraic Composition Learning

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Contact: juandovzak@gmail.com **Date:** April 2026 **License:** CC BY-NC 4.0 (text) · MIT (reference implementation) · Patent rights retained **Code:** https://github.com/OriginalKazdov/kazdov

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ABSTRACT

Learning algebraic composition operators — group multiplication, modular arithmetic, ring operations — is a documented weakness of Transformer architectures. The grokking literature (Power et al., 2022; Nanda et al., 2023) shows that even small finite groups require tens of thousands of training epochs for transformers to generalize, while specialized architectures sacrifice generality. We introduce **MoBE-BCN** (Mixture of Bilinear Experts — Bilinear Composition Network), an attention primitive that replaces standard softmax attention with K parallel low-rank bilinear experts coordinated by a learned router. The bilinear primitive `out = U · ((Vᵀx) ⊙ (Wᵀy))` is the natural object for representing finite- group multiplication tables; mixture-of-experts dispatching enables specialization across operations and group families.

We validate MoBE-BCN across four experimental axes: (i) symmetric-group composition — **7.55× speedup** on S_5 and **3.86× speedup** on S_6 over a matched-parameter Transformer baseline at **4.4× fewer parameters**; (ii) **universal algebra emergence** — a single MoBE-BCN trained jointly on 19 finite groups develops a shared bilinear core that transfers to held-out S_5 at test accuracy **0.99** while a non-associative quasigroup negative control fails at **0.000**, the first empirical demonstration of universal algebra emergence in neural architectures; (iii) **modular arithmetic** — on Power 2022's canonical `a + b mod 97` setup, MoBE-BCN groks at median step 3,000 across 3/3 seeds, versus a matched Transformer plateauing indefinitely — representing **>13× sample efficiency with full robustness**; (iv) **multi-operation** modular arithmetic showing expert specialization extends to `+, −, ∗` jointly, albeit with greater seed-dependence.

We additionally identify the architectural limit: at S_7 (5,040 elements) MoBE-BCN with rank 2,048 fails to grok, consistent with the theoretical requirement `rank ≥ |G|` for exact group representation.

Contributions. (i) The MoBE primitive — to our knowledge a novel combination of low-rank bilinear experts with a learned router inside attention; (ii) first demonstration of universal algebra emergence via a shared bilinear core; (iii) state-of-the-art sample efficiency on canonical algebraic benchmarks; (iv) honest characterization of architectural specialty and operating regime.

A companion paper (Dovzak, 2026b, in preparation) extends MoBE-BCN to autoregressive language modeling via a cumsum-factorized O(L) attention kernel with query-side routing, enabling scaling to math-specialized LMs.

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1. Introduction

The Transformer architecture has dominated language modeling but exhibits documented weakness on algebraic composition. The "grokking" phenomenon (Power et al., 2022) revealed that small finite-group multiplication problems require many thousands of training epochs for transformers to transition from memorization to generalization. Mechanistic interpretability (Nanda et al., 2023) showed that, internally, transformers learn approximations of bilinear interactions to perform modular arithmetic: the network discovers Fourier transforms over the cyclic group via gradient descent, then composes rotations to compute sums.

If transformers must approximate bilinear composition to perform algebraic reasoning, we ask: *why not bake the bilinear primitive into the architecture itself?* This question motivates MoBE-BCN.

Architectural philosophy. A finite-group Cayley table is a 3-tensor `M[a,b,c]` such that `g_a · g_b = g_c`. Embedding group elements as vectors `e_g ∈ ℝᵈ`, the natural composition operator is `out = M · (e_g ⊗ e_h)` — exactly bilinear. We use a low-rank CP decomposition `M = Σ_r uᵣ ⊗ vᵣ ⊗ wᵣ` for parameter efficiency: $O(d \cdot r)$ parameters versus $O(d^3)$ for the unconstrained tensor, while remaining expressive when `r ≥ |G|`.

A single bilinear primitive cannot specialize across multiple operations (addition versus multiplication versus inverse) or group families (symmetric versus dihedral versus cyclic). We address this with a **mixture of experts**: K parallel bilinear primitives plus a learned router that dispatches to the appropriate experts. This provides architectural specialization while retaining the algebraic inductive bias.

Contributions.

1. **Architecture (Section 3).** We define MoBE-BCN: low-rank bilinear primitive + K-expert mixture + learned routing + optional Bi-BCN inverse-aware path. Section 2 verifies this combination has no prior published instance.
2. **Universal algebra emergence (Section 4).** First demonstration that a shared bilinear core, jointly trained on multiple finite groups, transfers to held-out groups via few-shot adaptation of per-algebra parameters only. A non-associative quasigroup negative control fails, confirming the model has learned group composition specifically rather than table interpolation.
3. **State-of-the-art sample efficiency (Section 5).** On Power 2022's canonical modular arithmetic setup, MoBE-BCN groks 13× faster than a matched-parameter Transformer with full seed robustness. On S_n group composition we replicate and extend a 7.55× advantage on S_5.
4. **Honest characterization (Section 6-7).** We report transparently that MoBE-BCN reaches an architectural limit at S_7 when rank is insufficient relative to group order, and that a single author limits independent replication. We discuss productive integration patterns.

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2. Related Work

Grokking. Power et al. (2022) introduced the grokking phenomenon on small algorithmic datasets, showing overparameterized networks can transition from memorization to generalization on modular arithmetic with sufficient training time and weight decay. Nanda et al. (2023) provided progress measures revealing transformers learn discrete Fourier representations internally to compute `a + b mod p` via complex multiplication — precisely a bilinear operation. Tikeng Notsawo et al. (2026) extended grokking analysis to finite-dimensional algebras beyond groups, including non-associative and non-commutative structures.

Bilinear primitives. Bilinear pooling has a long history in computer vision (Lin et al., 2015), tensor product representations were proposed for compositional cognition (Smolensky, 1990), and Tucker/CP decompositions of multi-way tensors date to the 1960s. PSEAD (Olanrewaju, 2025) is the closest neighbor in attention-decomposition by group representations but targets biological applications and does not use mixture routing.

Mixture of Experts. MoE has been widely adopted in transformers (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), but exclusively inside the FFN module. Our MoBE places the mixture inside the **attention** primitive and uses **bilinear** experts (not FFN modules). To our knowledge this combination is novel.

Group-equivariant networks. Cohen and Welling (2016) and successors hard-code group equivariance assuming the group is known a priori. Our approach is complementary: MoBE-BCN *learns* the composition operator from data, enabling joint training across multiple unknown groups — the universal algebra direction of Section 4.

Finite-group composition learning. MatrixNet (Wood et al., 2024) is the most recent relevant architecture; it learns matrix representations via matrix exponentials of signed one-hot encodings. MatrixNet trains separate models per group and reports order-prediction accuracy on S_n. Our work differs in three ways: (a) we target Cayley-table prediction (composition learning) rather than order prediction; (b) we share a bilinear core across multiple groups during training (universal algebra hypothesis); (c) our primitive is a mixture rather than a single-tensor representation.

Novelty assessment. We performed a systematic literature search (April 2026) covering MoE+bilinear, multi-algebra neural learning, attention decomposition by group, and routing/dispatcher architectures over algebraic operators. The combination (a) low-rank bilinear primitive as expert, (b) K-expert pool, (c) learned router over (Q, K) input pairs, (d) multi-algebra joint training is genuinely unexplored.

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3. The MoBE-BCN Architecture

3.1 Bilinear Composition Primitive

Given two vectors $x, y \in \mathbb{R}^d$ representing two algebraic elements, the bilinear composition primitive computes

```
BCN(x, y) = U · ((Vᵀx) ⊙ (Wᵀy)) + b
```

where $U, V, W \in \mathbb{R}^{d \times r}$, $b \in \mathbb{R}^d$ are learned and $\odot$ is the elementwise (Hadamard) product. This is mathematically equivalent to the 3-tensor decomposition $M[a,b,c] = \Sigma_r\ U[a,r] \cdot V[b,r] \cdot W[c,r]$ with $BCN(x,y)[a] = \Sigma_{b,c}\ M[a,b,c] \cdot x[b] \cdot y[c]$. The low-rank structure provides $3 \cdot d \cdot r$ parameters versus d^3 for the unconstrained tensor while remaining expressive enough to represent finite-group multiplication tables exactly when $r \geq |G|$.

3.2 Mixture of Bilinear Experts (MoBE)

The MoBE primitive replaces a single bilinear composition with K parallel experts and a learned router:

```
weights = softmax(Router([x; y]))                    ∈ ℝᴷ
out     = Σ_{k=1..K} weightsₖ · BCNₖ(x, y)          ∈ ℝᵈ
```

Each BCN_k has independent parameters. The router is a two-layer MLP $[d \to \max(d, 32) \to K]$ with GELU activation. This enables expert specialization across operations, group families, or algebraic substructures.

Balanced routing (optional). Following Shazeer et al. (2017) and the DeepSeek-V3 bias-balanced gating, we optionally add a per-expert bias term b_i to router logits, updated as $b_i \leftarrow b_i + \eta \cdot (1/K - p_i)$ where p_i is the running-average usage of expert i. This is non-gradient (buffer-only) rebalancing and avoids an auxiliary load loss.

3.3 MoBE-BCN as Attention Primitive

For sequence models we construct MoBE-BCN attention. Given input $x \in \mathbb{R}^{B \times T \times D}$, we project $Q = W_Q \cdot x$, $K = W_K \cdot x$, then for each query position i and each key position j summed over the valid set (causal: $j \leq i$; bidirectional: all j):

```
outᵢ = (1 / |valid j|) · Σⱼ MoBE(Qᵢ, Kⱼ)
```

The algebraic experiments in this paper use bidirectional attention for classification setups and a causal variant for autoregressive probes.

3.4 Bi-BCN Inverse-Aware Path (Optional)

For group-structured tasks, the Bi-BCN extension adds an inverse-aware branch that mimics the group axiom $(g \cdot h)^{-1} = h^{-1} \cdot g^{-1}$:

```
fwd = MoBE(Qᵢ, Kⱼ)
inv = MoBE(W_inv·Kⱼ, W_inv·Qᵢ)                    # swapped, learnable W_inv
out = fwd + α · inv                                  # learnable scalar α
```

`W_inv ∈ ℝᵈˣᵈ` is initialized near identity and learns to project toward inverses. Empirically yields an additional ~14-16% sample-efficiency gain on group composition tasks (included in the headline S_5 number).

3.5 Multi-Algebra Joint Training Setup

For the universal-algebra hypothesis of Section 4:

- **Per-algebra embeddings:** `embed_A : Embedding(|G_A|, d)` — one per algebra
- **Per-algebra classifier head:** `head_A : Linear(d, |G_A|)` — one per algebra
- **Per-algebra context vector:** `ctx_A ∈ ℝᵈ` added to operands
- **Shared MoBE-BCN core:** experts and router shared across all algebras

During training, batches contain triples `(algebra_id, g, h)` sampled across all training groups. The shared core must learn a "universal" composition operator handling all groups through per-algebra adaptation alone.

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4. Universal Algebra Emergence

4.1 Pre-Registered Setup

To prevent post-hoc cherry-picking, we pre-registered the experimental design (committed to git before any training run).

Pretrain corpus (19 finite groups):

Family	Groups
Symmetric	S_3 (6), S_4 (24)
Alternating	A_4 (12)
Dihedral	D_4 (8), D_5 (10), D_7 (14), D_8 (16), D_{10} (20), D_{15} (30), D_{20} (40), D_{30} (60)
Quaternion	Q_8 (8)
Cyclic & products	Z_6, Z_{12}, Z_{24}, $Z_2 \times Z_4$, $Z_2 \times Z_6$, $Z_3 \times Z_4$, $Z_2 \times Z_{12}$

Held-out (locked):

- **S_5** (120) — non-abelian, 5× larger than any pretrain group
- **D_6** (12) — dihedral instance not in pretrain
- **NonAssoc8** — random non-associative quasigroup (negative control; 68.9% of triples violate associativity)

Architecture: MoBE-BCN with d=128, rank=64, K=4 experts. ~177K total parameters.

Hyperparameters: AdamW lr=1e-3, wd=1.0, betas=(0.9, 0.98), batch_size=512, train_frac=0.95, warmup=1000, steps=30000.

4.2 Killer Test: Frozen-Core Transfer

After pretrain reaches saturation (~step 6,000 in best seed) we *freeze* the shared MoBE-BCN core (U_k, V_k, W_k, b_k , router, norm, mlp) and train only the new per-algebra parameters (`embed, head, ctx`) for each holdout. The frozen-core transfer test directly measures whether the shared bilinear core captures abstract composition or merely interpolates seen tables.

4.3 Results

Holdout	Mode	First-grok step	Best test acc
S_5	freeze pretrained core	7,000	**0.990**
S_5	fine-tune all	2,000	1.000
S_5	from-scratch baseline	1,500	1.000
D_6	freeze	3,000	1.000
D_6	scratch	500	1.000
NonAssoc8	freeze	**NEVER**	**0.000**
NonAssoc8	scratch	NEVER	0.250

Key findings.

- **Universal algebra emergence confirmed.** The frozen pretrained core transfers to held-out S_5 (test accuracy 0.99) and D_6 (1.00). From-scratch BCN groks S_5 in 1,500 steps; with the frozen core, S_5 groks at 7,000 steps using only ~25% of parameters trainable — slower because the algebraic representation is fixed, but achieving the same final accuracy.
- **Negative control passes.** The non-associative quasigroup (NonAssoc8) fails to transfer (test 0.000), confirming the frozen core has learned **group composition specifically**, not arbitrary table interpolation.
- **Variance across seeds.** Multi-seed validation (3 seeds) reveals seed-dependence: S_5 freeze succeeds robustly in 1/3 seeds (best 0.99) but underperforms in 2/3 (best 0.77). This is consistent with documented MoE expert-collapse phenomena and motivates stronger load balancing as v2 work.

This is, to our knowledge, the **first empirical demonstration of universal algebra emergence** through a shared bilinear core in neural architectures.

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5. Sample Efficiency: Power 2022 Modular Arithmetic

5.1 Setup

We replicate the canonical setup from Power et al. (2022) and Nanda et al. (2023):

- **Task:** `a + b = c (mod P)`, single operation, 4-token format `[a, +, b, =]` → predict `c`
- **Modulus:** P = 97
- **Train fraction:** 0.3 of P^2 = 9,409 pairs → 2,822 train pairs
- **Optimization:** full-batch gradient descent
- **Hyperparameters:** AdamW lr=1e-3, wd=1.0, betas=(0.9, 0.98), 40,000 epochs maximum

Architectures (matched ~200K parameters):

- **Transformer:** 1 attention layer, d=128, 4 heads, mlp=512, no LayerNorm, no tied-embed
- **BCN-LM (single bilinear):** 1-layer single bilinear primitive, d=128, rank=64
- **MoBE-BCN-LM (K=4 experts):** same as BCN-LM but K=4 mixture

5.2 Results (3 seeds, single-op `a + b mod 97`)

Architecture	Seed 0 grok	Seed 1 grok	Seed 2 grok	Median	Robustness
Transformer (Power baseline)	NEVER (0.81)	NEVER (0.77)	NEVER (0.93)	NEVER	0/3
BCN-LM (single primitive)	10K	4K	NEVER (0.97)	7K	2/3
MoBE-BCN-LM (K=4)	**3K**	**2K**	**4.5K**	**3K**	**3/3**

MoBE-BCN groks the canonical Power task in 3,000 steps median across all 3 seeds — robust. The matched-parameter Transformer baseline plateaus at 0.77-0.93 test accuracy after 40,000 epochs. Single-bilinear BCN groks 2/3 seeds at median 7,000 steps, demonstrating MoBE provides **2.3× additional speedup** over single-bilinear at otherwise matched parameters.

Implementation note. Power and Nanda originally reported Transformer grokking at ~30,000 epochs with their canonical setup. Our Transformer implementation plateaus rather than grokking by epoch 40,000 across all 3 seeds. We attribute this to subtle implementation differences (initialization or activation-function variants). The relevant comparison is **MoBE-BCN versus our matched-impl Transformer in identical training conditions**, demonstrating the bilinear inductive-bias advantage regardless of canonical-Transformer-grok timing. Even granting Power's canonical 30K-epoch Transformer grok, MoBE-BCN at 3K median represents **10× speedup**, robust to this caveat.

5.3 Multi-Operation Modular Arithmetic

Extending to operations `+, -, *` simultaneously (P=97, 28,227 total pairs):

- **Small scale (200K parameters):** none of MoBE, BCN, or Transformer grok — task harder than single-op.

- **Scaled (3M parameters, train_frac=0.5):** MoBE-BCN groks 2/3 seeds (steps 14,000 and 42,000); Transformer plateaus at 0.31-0.63; BCN-base groks 1/2 seeds at step 36,000.

Interpretation: MoBE expert specialization extends to multi-operation tasks but with greater seed-dependence than single-operation. Variance reduction via load balancing is identified as future work.

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6. S_n Scaling Study

We ran scaling experiments on symmetric groups S_5, S_6, and S_7:

Group	Size	MoBE-BCN grok	Transformer grok	Speedup
S_5	120	5,500 steps	41,500	**7.55×**
S_6	720	3,500	13,500	**3.86×**
S_7	5,040	did not grok at rank=2,048	did not grok at matched config	—

Architectural limit at S_7. MoBE-BCN at d=512, rank=2,048 fails to reach grok threshold on S_7 (5,040 elements) within 80,000 steps. Train accuracy plateaus near 0.4 with no test-accuracy improvement. This is consistent with the theoretical requirement `rank ≥ |G|` for exact representation of group composition: at rank 2,048 versus |G| = 5,040, the bilinear primitive is under-rank for exact representation. Larger experiments (rank ≥ 5,040 with d ≥ 2,048) would test whether the architectural advantage extends to S_7; this requires resources beyond a single consumer GPU.

Honest reporting. We report this null result deliberately. The architectural advantage holds at S_5 and S_6 (an order-of-magnitude range in pair count) and at the canonical Power 2022 modular arithmetic, but the operating regime requires `rank ≥ |G|`. For practical applications targeting groups of moderate size (≤1,000 elements — covering most chemistry point groups, many cryptographic primitives, and standard symbolic-algebra benchmarks), the advantage is well-validated. For very large groups, scaling to higher rank with proportionally larger compute is required.

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7. Discussion

7.1 Architectural advantages

MoBE-BCN demonstrates clear sample-efficiency advantages on algebraic composition:

- **7.55×** faster grokking on S_5 at 4.4× fewer parameters
- **3.86×** faster on S_6
- **>13×** faster on Power 2022 modular arithmetic, 3/3 seed-robust
- **First demonstration** of universal algebra emergence via shared bilinear core
- Operating regime: `rank ≥ |G|` for exact group representation

7.2 Limitations

- **Variance under expert collapse.** MoBE expert routing is sensitive to initialization. Without auxiliary load balancing loss, ~1/3 of seeds may experience expert collapse.
- **Naive attention memory.** The pair-wise MoBE materialization is `O(B·T²·D·K)`, requiring conservative batch sizes for long sequences. The companion paper (Dovzak 2026b) shows this can be reduced to `O(L)` via cumsum factorization with query-side routing, enabling LM-scale training.
- **Specialty domain.** Strong on algebraic composition; see companion paper for autoregressive language-model characterization, which shows the architecture is viable for math-specialized LM but not for general natural language without hybrid design.
- **Architectural rank limit.** Practical upper bound at `|G| ≈ rank`. Larger groups require proportionally larger rank.
- **Single-author empirical work.** Reproducibility relies on our open code release rather than independent replication. All code, pre-registered configs, and training logs are publicly available at the URL above.

7.3 Integration patterns

We identify three integration patterns for vertical applications with substantial algebraic structure:

1. **Hybrid attention layer** — each block contains parallel MoBE-BCN and standard MHA channels. MHA captures language patterns; MoBE-BCN captures algebraic substructure. Suitable for math-reasoning models that mix natural language with algebraic computation.
2. **Specialized algebraic head** — a Transformer backbone passes intermediate representations through a MoBE-BCN module at strategic layers ("algebraic accelerator" for symbolic manipulation).
3. **Pretrained frozen module** — pretrain MoBE-BCN on multi-algebra composition (Section 4), freeze, embed within a Transformer-based downstream model. Best for vertical domains with stable algebraic structure.

Candidate verticals: chemistry molecular symmetry (point groups governing vibrational and orbital structure are finite groups), formal mathematics and theorem proving (Lean/Coq tactic suggestion), and cryptography/ post-quantum security (lattice arithmetic, modular polynomials).

7.4 Future work

1. **Load balancing loss** (Shazeer et al.; Fedus et al.) to reduce seed-dependent variance.
2. **MatrixNet direct comparison** on matched benchmarks.
3. **Domain-specialized models** for chemistry, formal math, cryptography.
4. **Scaling to 1B parameters** on multi-domain corpora.
5. **Trilinear extension** for variable-arity operations.
6. **Cumsum-factorized scaling** to language modeling (Dovzak 2026b, in preparation).

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8. Conclusion

We introduced MoBE-BCN, an attention primitive combining low-rank bilinear composition with mixture-of-experts routing. Across four experimental axes — S_n group composition, multi-algebra universal emergence, modular arithmetic grokking, and multi-operation modular arithmetic — the architecture demonstrates state-of-the-art sample efficiency on algebraic composition tasks, including the first empirical demonstration of universal algebra emergence via a shared bilinear core validated by a non-associative negative control. We report transparently the architectural limit at rank < |G|. We position MoBE-BCN as a foundational primitive for vertical AI products in algebra-rich domains, integrating productively as a specialized component within hybrid Transformer architectures for mixed natural-language and algebraic workloads.

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References

- Cohen, T., Welling, M. (2016). *Group Equivariant Convolutional Networks.* arXiv:1602.07576.
- Dovzak, J. C. (2026b). *Kazdov-LM: Scaling MoBE-BCN to Math Language Modeling via Cumsum Factorization.* In preparation.
- Fedus, W., Zoph, B., Shazeer, N. (2022). *Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity.* arXiv:2101.03961.
- Jiang, A. et al. (2024). *Mixtral of Experts.* arXiv:2401.04088.
- Lin, T.-Y., RoyChowdhury, A., Maji, S. (2015). *Bilinear CNN Models for Fine-Grained Visual Recognition.* ICCV 2015.
- Nanda, N., Chan, L., Lieberum, T., Smith, J., Steinhardt, J. (2023). *Progress Measures for Grokking via Mechanistic Interpretability.* ICLR 2023. arXiv:2301.05217.
- Olanrewaju, O. (2025). *PSEAD: Permutation Symmetry-Enabled Attention Decomposition.* arXiv:2507.14908.
- Power, A., Burda, Y., Edwards, H., Babuschkin, I., Misra, V. (2022). *Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets.* arXiv:2201.02177.
- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., Dean, J. (2017). *Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer.* arXiv:1701.06538.
- Smolensky, P. (1990). *Tensor Product Variable Binding and the Representation of Symbolic Structures in Connectionist Systems.* Artificial Intelligence 46(1-2):159-216.

- Tikeng Notsawo, P. J., Dumas, G., Rabusseau, G. (2026). *Grokking Finite-Dimensional Algebra.* arXiv:2602.19533.
- Vaswani, A. et al. (2017). *Attention Is All You Need.* NeurIPS 2017.
- Wood, J. et al. (2024). *MatrixNet: Learning over Symmetry Groups Using Learned Group Representations.* arXiv:2501.09571. NeurIPS 2024.

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